

Particulate Matter Monitoring & Health Effects

Issue 2026-09-27 | Entry window 27 September 2026 (full day, Sunday) | 4 records screened in scope

4

RECORDS IN SCOPE
(7 EXCLUDED)

3

OF 10 SUBTOPIC
BANDS POPULATED

2

EFFECT ESTIMATES
WITH A CI

1

TIER-A
RECORDS

1

RECORDS CARRIED ON
METADATA ALONE

Signal of the day

A personal-monitor panel in older adults finds a next-day activity deficit after higher-than-usual PM exposure — and no association at all when the same days are scored with area-level PM.

Bloomberg and colleagues (*medRxiv*, tier A, not peer reviewed) had **67 UK adults aged 50–85** wear personal air-quality monitors and 24-h accelerometers for five days (**304 person-days**). Within-person, a day **+1 IQR above the person's own mean** — only **+2.6 $\mu\text{g m}^{-3}$ PM_{2.5}** — was followed by **8 min less light activity** (95% CI **−14 to −2**) and **483 fewer steps** (**−855 to −82**); PM₁₀ gave the same LPA deficit and **8 min more sedentary time**. MVPA did not move, and **area-level PM estimates produced no association**. **Why it is the signal:** the contrast is small enough to sit inside the error band of an uncalibrated optical sensor, yet it is the personal series, not the modelled area series, that carries the within-person signal — the pattern expected if area estimates smear the day-to-day variance that behaviour responds to. The weaknesses: monitor model and calibration are not stated, weather and minor illness drive both indoor PM and next-day activity, and the reverse pathway (being indoors more raises personal PM) is only partly handled by the next-day lag. **The operational rule:** in short-term panel designs, the personal sensor is not a noisier version of the area estimate but a different exposure; report both, and report the sensor's co-location performance so the contrast can be judged against it.

What changed today

- **Sunday window, thin by construction.** PubMed returned **4** health records and **0** on the sensing axis; Europe PMC **1**, Crossref-ISSN **2**, arXiv **1** (off-topic engine control). Four records were admitted.
- **Bangladesh's regulatory network reports a 15% decline.** Shahadat et al. (16 DoE stations, 2018–2025): network PM_{2.5} **91.54 → 77.64 $\mu\text{g m}^{-3}$** , PM₁₀ **155.7 → 132.3**, non-monotonic after the 2020 minimum, every station above national standards every year, winter/monsoon ratio **~4.2** for PM_{2.5}. Kriging was abandoned for prediction because 16 stations cannot support it.
- **NAKO mental-health results arrived without an abstract.** The German National Cohort longitudinal environmental-exposure paper (*Environ Int*) is carried metadata-only and goes to the top of the retry list.
- **Bioaerosol in a paediatric emergency room.** 73.47% of culturable bacterial aerosol was below 3.3 μm , and CFD put 79.71% of deposition on the desk and floor — a fraction optical PM sensors cannot distinguish.

Window continuity

The 26 September issue (built earlier on this run) closed with `last_entry_date` at 26 September. This issue collects **27 September in full**, continuous with it. Built on the **29 September backfill run**; **28 September** follows as its own issue on the same run, and **29 September is not opened** under the 22:00 rule. Sunday is not a weekly trigger under the Saturday rule: the W39 weekly closed on 26 September.

Corpus analytics

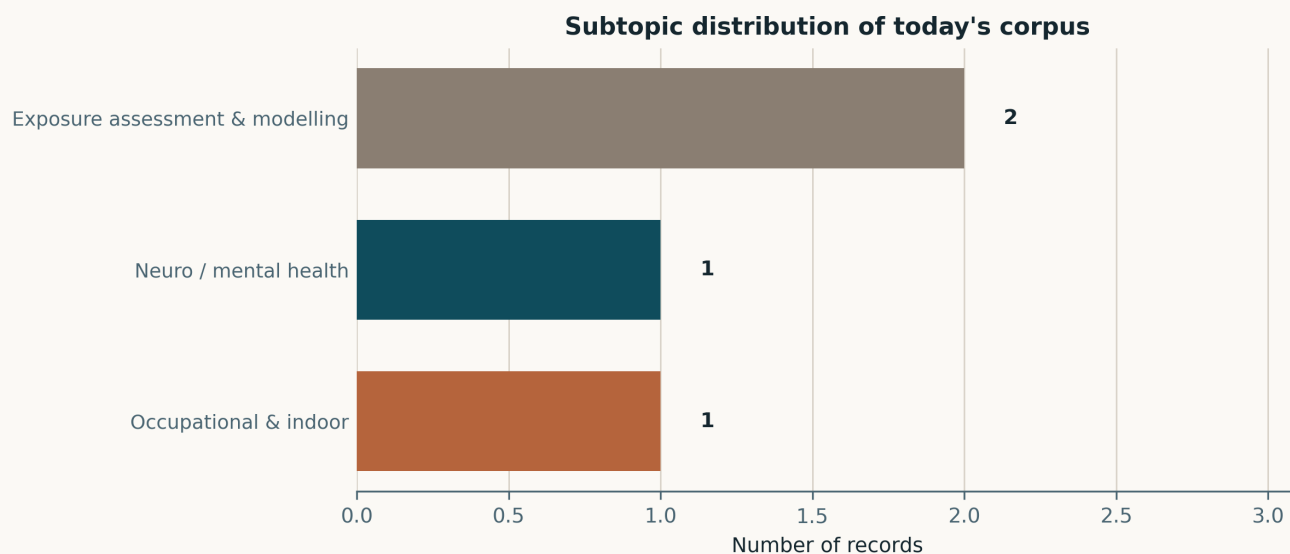


Fig. Subtopic assignment of the four in-scope records: *Exposure assessment & modelling* **two** (UK personal-monitor panel, Bangladesh network), *Neuro / mental health* and *Occupational & indoor* **one** each. Seven bands are empty.

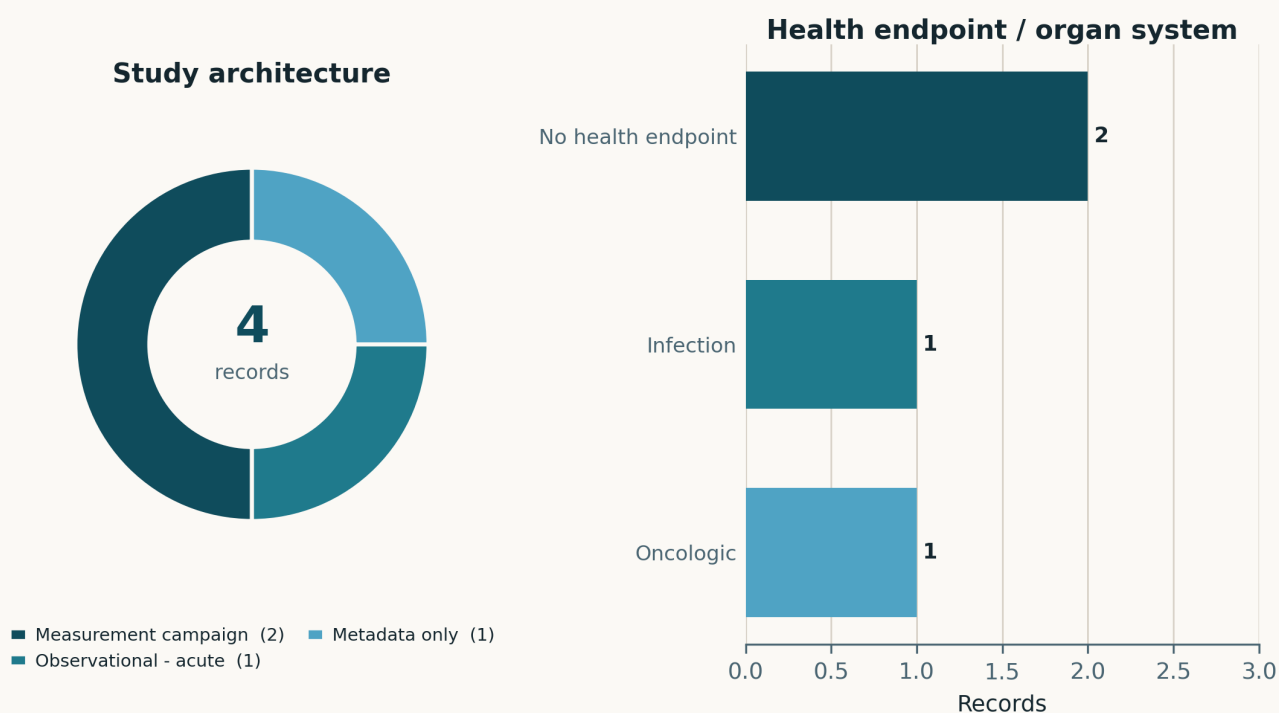


Fig. Left — architecture: two measurement campaigns (Bangladesh network, hospital bioaerosol), one acute observational panel, one metadata-only deposit. Right — endpoint: *no health endpoint* **two** (the behaviour panel and the metadata-only record), *infection* one, and *oncologic* one (the Bangladesh ELCR screen).

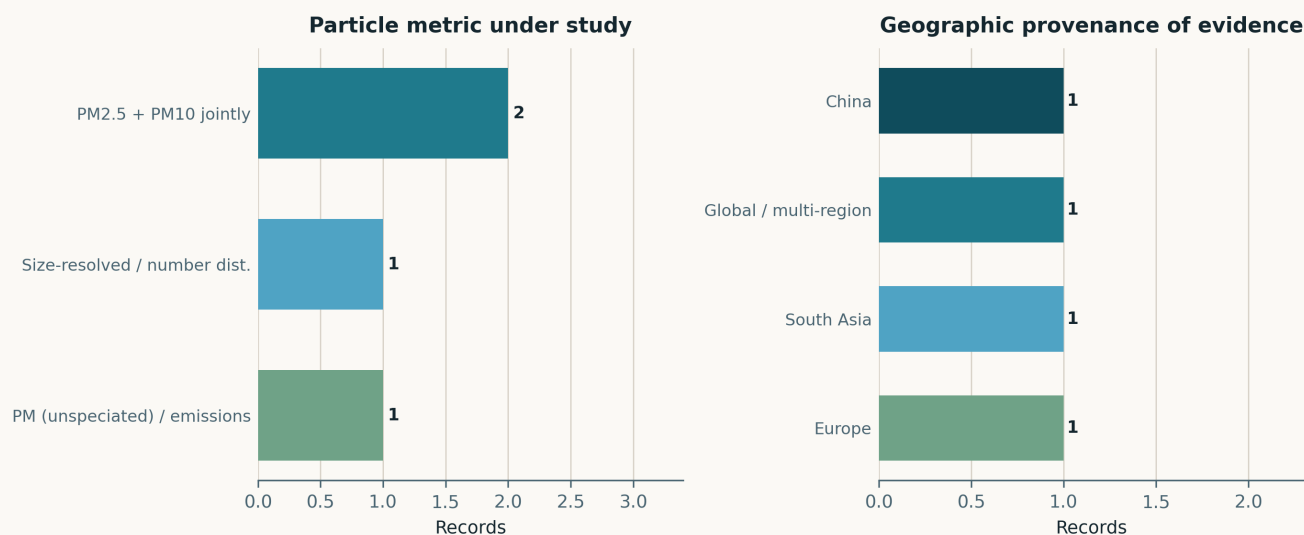


Fig. Left — particle metric: PM_{2.5} + PM₁₀ **two**, size-resolved bioaerosol **one**, unspeciated **one**. Right — geography: UK, Bangladesh, China, and one metadata-only record binned as not recoverable.

Life-course windows addressed today (records may span several stages)

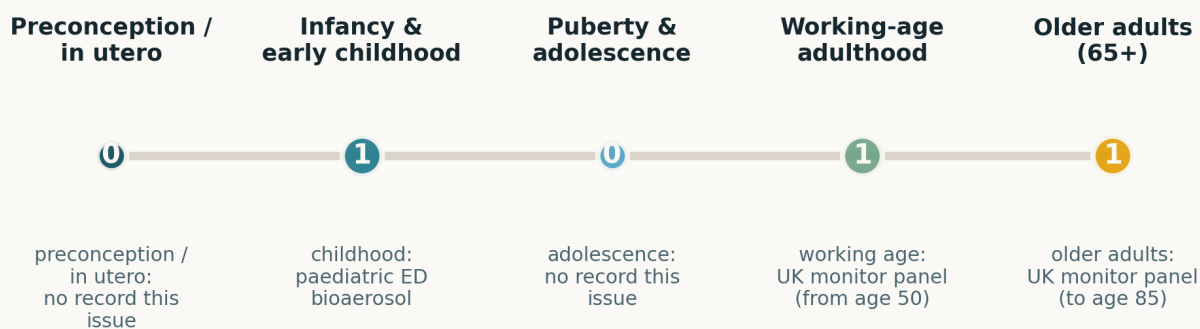


Fig. Life-course windows: the UK panel spans working age and older adults (50–85) and is counted in both; the paediatric ED study counts in childhood.

Where the work is happening: subtopic x study architecture

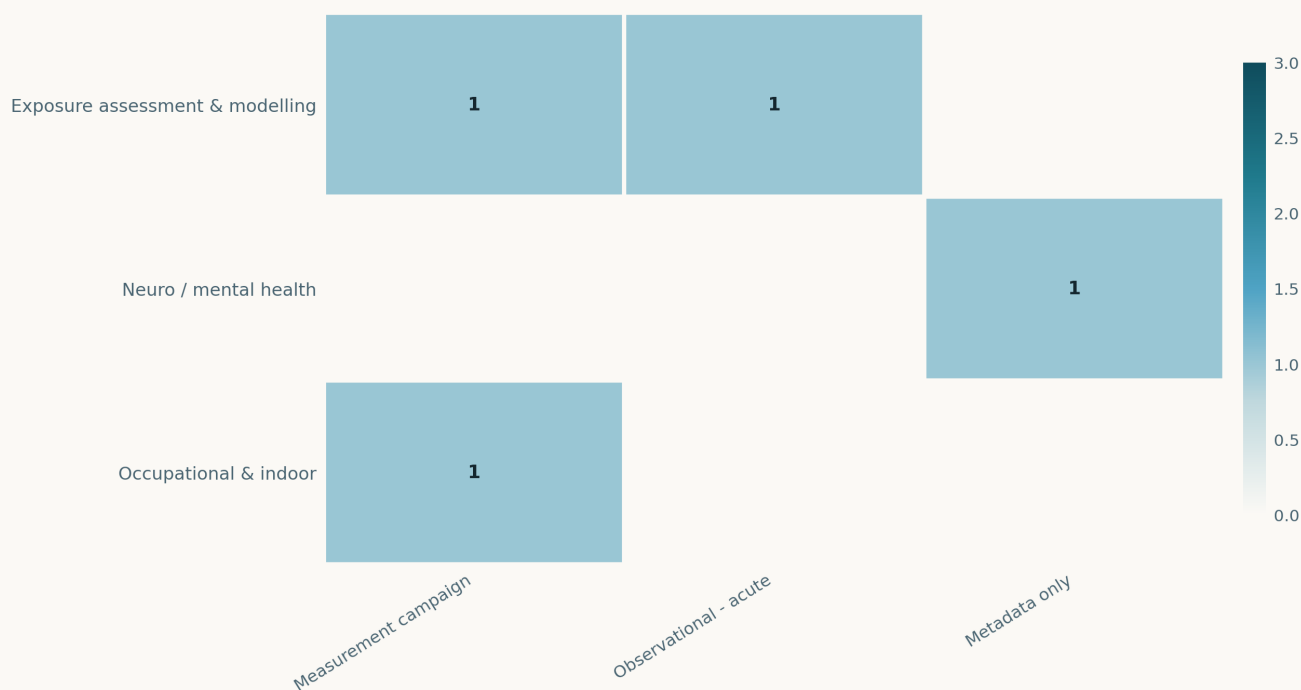


Fig. Subtopic × study architecture for the four records; four occupied cells, each a singleton; *Exposure* splits between measurement campaign and acute observational.

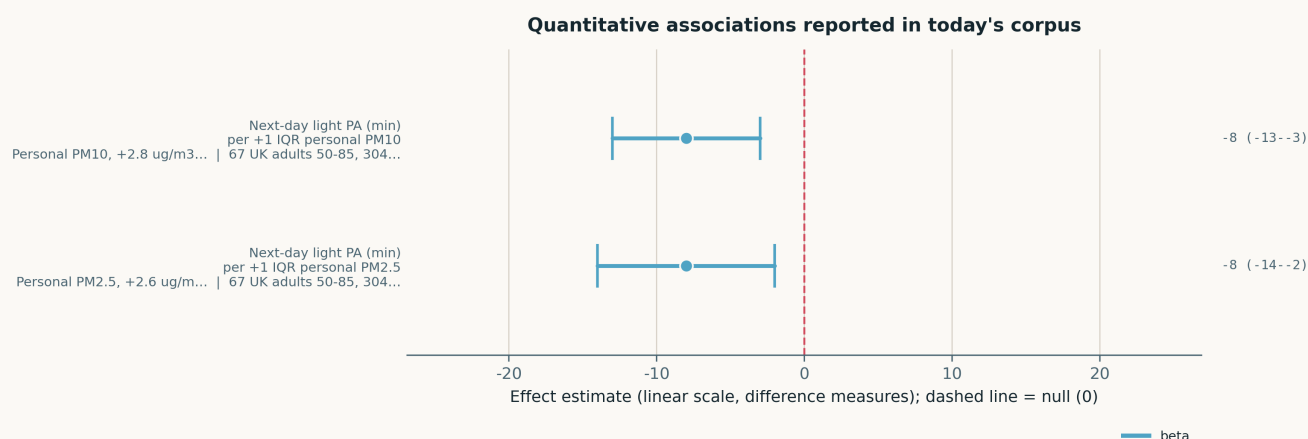


Fig. Two difference-scale estimates from one preprint. Next-day light physical activity, minutes, per +1 IQR within-person deviation in personal PM_{2.5} (−8, −14 to −2) and PM₁₀ (−8, −13 to −3). Same 67 participants and days; not independent. Step-count estimates are on a different scale and are given in the digest text.

Paper-level digest

Ordered by cluster. Each entry gives the methodological core and the finding that matters, not an abstract paraphrase. Records marked *metadata only* carry no recoverable abstract and assert no finding.

Neuro / mental health

1 record

Wang et al. 2026 Environ Int

Environmental exposures and mental health: longitudinal results from the German National Cohort (NAKO)

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 27 Sep with no abstract; Crossref, Europe PMC and the publisher landing page returned nothing. The title indicates longitudinal analyses of environmental exposures and mental health in the German National Cohort; which exposures (air pollution, noise, greenness, heat) are included cannot be stated. NAKO's size (~200,000) makes it a priority retry. Nothing about exposure metric, sample size or estimates can be stated from the deposit, and this watch does not infer a method from a title.

doi: 10.1016/j.envint.2026.110539

Exposure assessment & modelling

2 records

Bloomberg et al. 2026 medRxiv (preprint)

Associations of individual exposure to particulate matter with next-day physical activity and sedentary behaviour in older adults: a micro-longitudinal study

Micro-longitudinal preprint: 67 UK adults aged 50-85 wore personal air-quality monitors (waking hours) and 24-h accelerometers for five days (304 person-days). Within-person mixed models: a higher-than-usual day (+1 IQR, i.e. +2.6 $\mu\text{g m}^{-3}$ PM_{2.5} above the person's mean) was followed by 8 min less light activity (95% CI -14 to -2) and 483 fewer steps (-855 to -82); for PM₁₀ (+2.8 $\mu\text{g m}^{-3}$), 8 min less LPA (-13 to -3), 8 min more sedentary time (0 to 15) and 412 fewer steps. No MVPA association, and none with area-level PM. Not peer reviewed; the monitor model and any calibration are not stated; a 2.6 $\mu\text{g m}^{-3}$ contrast is within typical optical-sensor error, so residual weather or illness confounding of day-to-day behaviour is the main threat. Relevance: high - the association exists only in personal-sensor exposure, not area estimates.

doi: 10.64898/2026.09.23.26363783

Shahadat et al. 2026 Environ Geochem Health

Spatiotemporal variability, trends, and health risk assessment of ambient PM_{2.5} and PM₁₀ in Bangladesh during 2018-2025.

Analysis of 2018-2025 PM_{2.5} and PM₁₀ from the 16 Department of Environment continuous stations in Bangladesh, with seasonal Mann-Kendall trends and ELCR/HQ risk indices. Every station exceeded national standards and WHO guidelines every year. Network annual mean PM_{2.5} fell from 91.54 to 77.64 $\mu\text{g m}^{-3}$ (-15.2%) and PM₁₀ from 155.7 to 132.3 (-15.0%), non-monotonically, rebounding after the 2020 minimum; winter/monsoon ratios were ~4.2 (PM_{2.5}) and 3.0 (PM₁₀). Most seasonal trends were not significant. The authors note kriging performed poorly and was used only for display - 16 stations cannot map a country of 170 million. ELCR applied to PM_{2.5} mass is a screening construct, not an epidemiological risk. Relevance: the station-data quality control is undescribed, and this network is exactly what low-cost sensor densification in Bangladesh would calibrate against.

doi: 10.1007/s10653-026-03506-z

Occupational & indoor

1 record

Liu et al. 2026 Environ Res*Spatiotemporal Contamination Characteristics, Bacterial Community Structure, and Airflow-Mediated Transport of Bacterial Aerosols in a Pediatric Emergency Consulting Room.*

One paediatric emergency consulting room: culturable airborne and surface bacteria, high-throughput sequencing and CFD of aerosol transport and deposition. Airborne culturable counts differed little between locations (well-mixed small room), surface deposition was heterogeneous, and 73.47% of bacterial aerosol was below 3.3 μm ; *Staphylococcus* was the core genus. CFD placed the consultation desk and floor at 79.71% of deposited particles, with high-risk zones around occupants. One room, culturable fraction only (typically <1% of total bacteria), and sampling duration and counts are not in the abstract. Relevance: marginal; bioaerosol is invisible to optical PM sensors, so ventilation decisions here need culture or fluorescence methods.

doi: 10.1016/j.envres.2026.125789

Corpus register

First author	Focus	Journal	Design	Metric	Region
Neuro / mental health					
Wang et al.	Environmental exposures and mental health: longitudinal results from the German National Cohort	<i>Environ Int</i>	Metadata only (no abstract)	Metadata only - environmental exposures and mental health in NAKO (by title)	Not recoverable from metadata
Occupational & indoor					
Liu et al.	Spatiotemporal Contamination Characteristics, Bacterial Community Structure, and Airflow-Mediated Transport of Bacterial Aerosols in a Pediatric Emergency Consulting Room.	<i>Environ Res</i>	Repeated field measurement with CFD simulation	Size-resolved culturable bacterial aerosol (Andersen-type stages), CFD deposition	China (paediatric hospital)
Exposure assessment & modelling					
Bloomberg et al.	Associations of individual exposure to particulate matter with next-day physical activity and sedentary behaviour in older adults: a micro-longitudinal study	<i>medRxiv (preprint)</i>	Panel study	Personal air-quality monitors (waking hours, 5 days) vs area-level PM _{2.5} / PM ₁₀	UK
Shahadat et al.	Spatiotemporal variability, trends, and health risk assessment of ambient PM _{2.5} and PM ₁₀ in Bangladesh during 2018-2025.	<i>Environ Geochem Health</i>	Monitoring-record analysis	16 DoE continuous monitoring stations, PM _{2.5} and PM ₁₀ , 2018-2025	Bangladesh (national, 16 stations)

Provenance and method

Entry window **27 September 2026** (Sunday), single day, continuous with the 26 September issue. Harvester legs run leg-by-leg: **PubMed** health **4**, sensing **0**; **Europe PMC** **1**; **Crossref by ISSN** **2**; **arXiv** **1** entry dated 27 September (off-topic); **OpenAlex** **0**. The **PubMed connector** ([EDAT] 26–28 September, three axes, bucketed by entrez date) placed **6** PMIDs on this day, **5** appended. **11** candidates screened, **4** **admitted**, **7** **rejected** with logged reasons (one had already been rejected on 26 September). **Consensus** and **ClinicalTrials.gov** were swept once for the whole 26–28 September backfill and are reported in the 26 September issue (0 new records; one trial update). **Scholar Gateway** returned an identity error (Could not resolve user identity) and contributed nothing this run. One record carried on metadata alone. DOI gate: **0 fail**, **0 warn**. Abstracts remain the copyright of their respective publishers.